Barış Yılmaz
G191210303
Bilgisayar Mühendisliği - 2/B

B. Yılmaz

1) $r < R$ için

$$Q(r) = \int_0^r \rho(r)\, dV = \int_0^r \rho_0 \left(1 - \frac{R}{2r}\right) \cdot 4\pi r^2 dr$$

$$= 4\pi \rho_0 \int_0^r \left(1 - \frac{R}{2r}\right) r^2 dr$$

$$= 4\pi \rho_0 \int_0^r r^2 - \frac{Rr}{2}\, dr$$

$$= 4\pi \rho_0 \left(\int_0^r r^2 dr - \frac{R}{2} \int_0^r r\, dr \right)$$

$$= 4\pi \rho_0 \left(\frac{1}{3} r^3 \Big|_0^r - \frac{R}{2} \left(\frac{1}{2} r^2 \Big|_0^r \right) \right) = 4\pi \rho_0 \left(\frac{r^3}{3} - \frac{r^2 R}{4} \right)$$

$$E(r) = \frac{Q(r)}{\varepsilon_0 . A} \Rightarrow \frac{\cancel{4\pi} \rho_0 \cancel{r^2} \left(\frac{r}{3} - \frac{R}{4} \right)}{\cancel{4\pi r^2} . \varepsilon_0} = \frac{\rho_0 \left(\frac{r}{3} - \frac{R}{4} \right)}{\varepsilon_0} \quad //$$

$r > R$ için

$$Q(R) = Q_{iç} = 4\pi \rho_0 \left(\frac{R^3}{3} - \frac{R^2 R}{4} \right) = 4\pi \rho_0 \frac{R^3}{12}$$

$$E(r) = \frac{Q_{iç}}{\varepsilon_0 . A} = \frac{\cancel{4\pi} \rho_0 \frac{R^3}{12}}{\cancel{4\pi} r^2 \varepsilon_0} = \frac{\rho_0 \frac{R^3}{12}}{r^2 \varepsilon_0} = \frac{\rho_0 R^3}{12 r^2 \varepsilon_0} \quad //$$

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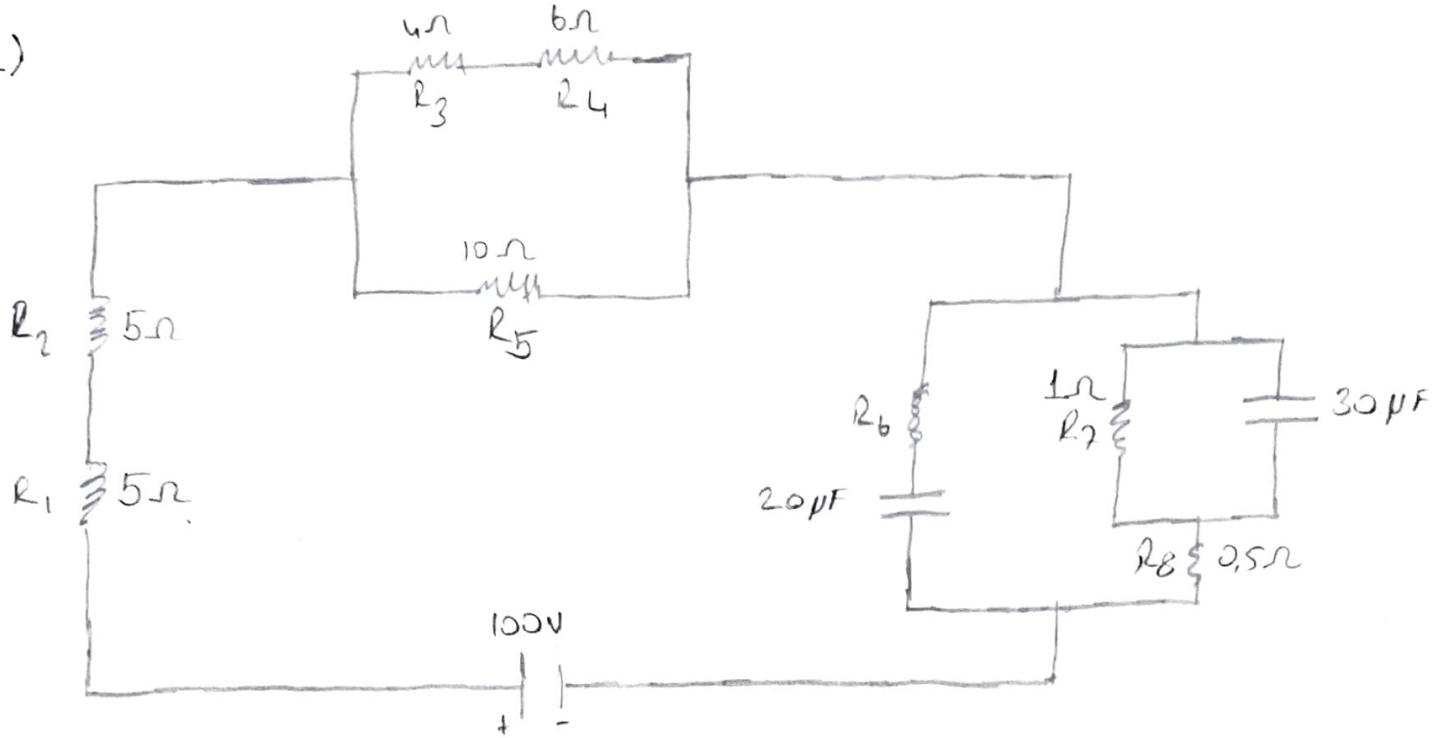
2)



$R_{es} = 5+5+5+1+0,5 = 16,5$

$V = R_{es}.I \Rightarrow 100 = 16,5.I \Rightarrow I = 6,06\ A$

$V_{AB} = R_7.I = 6,06.1 = 6,06\ V$

$V_{AC} = (R_7+R_8).I = 6,06.1,5 = 9,09\ V$

$Q = C.V \Rightarrow Q_1 = 30\mu F.\ 6,06 = 1,818.10^{-4}\ C$,,

$Q_2 = 20\mu F.\ 9,09 = 1,818.10^{-4}\ C$,,

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3) Manyetik alan içindeki yüke etki eden kuvvet $\boxed{\vec{F} = q\,(V \times B)}$

$$\boxed{\lambda = \frac{q}{2\pi R}}$$

$$dF = dq\,V B \sin Q$$
$$= \lambda\, dl\, V B \sin Q$$
$$= \frac{q}{2\pi \cancel{R}} \cancel{R}\, dQ\, V B \sin Q$$
$$= \frac{q}{2\pi}\, dQ\, V B \sin Q$$

$$\Rightarrow \int dF = F = \frac{qVB}{2\pi} \cdot \int \sin Q\, dQ$$

$$^1F \Rightarrow \frac{qVB}{2\pi} \int_{\pi}^{2\pi} \sin Q\, dQ = \frac{qVB}{2\pi} \cdot -\cos Q \Big|_{\pi}^{2\pi}$$
$$= \frac{qVB}{2\pi} \cdot \left((-\cos 2\pi) - (-\cos \pi)\right)$$
$$= \frac{qVB}{\cancel{2}\pi} \cdot -\cancel{2} = -\frac{qVB}{\pi} = -\frac{qwRB}{\pi}$$

$$^2F \Rightarrow \frac{qVB}{2\pi} \int_{0}^{\pi} \sin Q\, dQ = \frac{qVB}{2\pi} \cdot -\cos Q \Big|_{0}^{\pi}$$
$$= \frac{qVB}{2\pi} \cdot \left((-\cos \pi) - (-\cos 0)\right)$$
$$= \frac{qVB}{\cancel{2}\pi} \cdot \cancel{2} = \frac{qVB}{\pi} = \frac{qwRB}{\pi}$$

$$\tau = \left[\left|\frac{qwRB}{\pi}\right| \times R + \left|-\frac{qwRB}{\pi}\right| \times R\right]$$

$R \cdot \sin 90^\circ$ $\qquad$ $R \cdot \sin 270^\circ \Rightarrow \dfrac{2qwR^2B}{\pi}$ //

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4)

a) $\vec{\mu} = N.I\,\vec{A} \longrightarrow \pi r^2$

$= 15 . 7,6 . \pi . \left(\frac{0,22}{2}\right)^2 . (-\hat{k})$

$= (-4,334\hat{k})\,Am^2$ //

b) $\tau = \vec{\mu} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -4,334 \\ 0,55 & 0,60 & -0,65 \end{vmatrix}$

$= (-4,334\hat{k}\,A.m^2) \times (0,55\hat{i} + 0,60\hat{j} - 0,65\hat{k})\,T$

$= (2,6\hat{i} - 2,6\hat{j})\,mN$ //

c) $U = -\vec{\mu} . \vec{B}$

$= (-4,334\hat{k}\,Am^2) \times (0,55\hat{i} + 0,60\hat{j} - 0,65\hat{k})\,T$

$= -(4,334\,Am^2)(0,65T)$

$= -2,81\,J$ //

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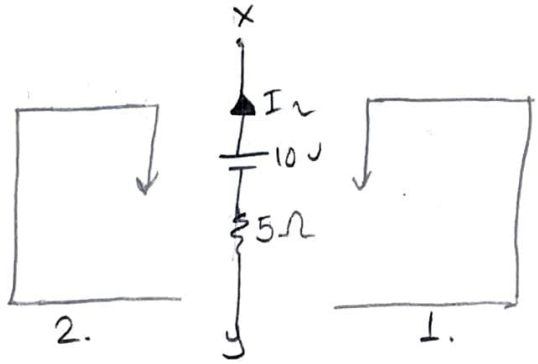
5)

a)



x için $\Rightarrow I_1 + I_2 - I_3 = 0$

1. çevrim $\Rightarrow 2I_3 - 20 + 3I_3 - 10 + 5I_2 = 0$

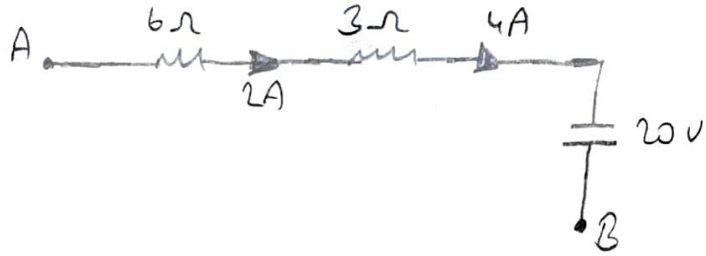
2. çevrim $\Rightarrow -4I_1 + 20 - 6I_1 - 10 + 5I_2 = 0$

$$\begin{aligned} I_1 + I_2 - I_3 &= 0 \\ I_2 + I_3 &= 6 \\ + \quad 2I_1 - I_2 &= 2 \\ \hline 3I_1 + I_2 &= 8 \end{aligned}$$

$$\begin{aligned} 3I_1 + I_2 &= 8 \\ + \quad 2I_1 - I_2 &= 2 \\ \hline 5I_1 &= 10 \end{aligned}$$

$\rightarrow I_1 = 2A \quad I_2 = 2A \quad I_3 = 4A$ //

b)



$V_A - 12 - 12 + 20 = V_B$

$V_{AB} = 4V$ //